

Computer Networks — Quick Revision Notes

1. Introduction to Computer Networks

A **computer network** is a collection of interconnected devices that communicate and share resources.

Advantages

- Resource sharing
 - Fast communication
 - Data sharing
 - Internet access
 - Cost reduction
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2. Applications of Computer Networks

Major Applications

- Resource sharing
- Communication (Email, Chat, Video calls)
- Online banking and e-commerce
- Education and e-learning
- Entertainment and streaming
- Business collaboration
- Cloud computing
- Remote access

Examples

- Google Meet
 - YouTube
 - Amazon
 - Google Drive
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3. Modes of Communication

(a) Simplex

One-way communication.

Examples:

- TV broadcasting

- Keyboard to computer

(b) Half Duplex

Two-way communication but one direction at a time.

Example:

- Walkie-talkie

(c) Full Duplex

Two-way communication simultaneously.

Examples:

- Mobile calls
- Video calls

Comparison

Mode	Direction	Example
Simplex	One-way	TV
Half Duplex	Both ways, one at a time	Walkie-talkie
Full Duplex	Both ways simultaneously	Phone call

4. LAN, MAN, WAN

LAN (Local Area Network)

- Covers small area
- High speed
- Example: School lab

MAN (Metropolitan Area Network)

- Covers a city
- Connects multiple LANs
- Example: City cable network

WAN (Wide Area Network)

- Covers large geographical area
- Example: Internet

Comparison

Feature	LAN	MAN	WAN
Area	Small	City	Large
Speed	High	Medium	Lower
Cost	Low	Medium	High

5. Internet

The Internet is a worldwide network connecting millions of devices.

Features

- Global connectivity
- Fast communication
- Information sharing
- 24×7 availability

Services

- WWW
- Email
- File transfer
- Online banking
- E-commerce

Important Terms

- ISP (Internet Service Provider)
 - IP Address
 - Browser
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6. Network Hardware

Important Devices

NIC

Connects computer to network.

Hub

Broadcasts data to all devices.

Switch

Sends data only to destination device.

Router

Connects different networks.

Modem

Converts digital and analog signals.

Transmission Media

Used for data transfer.

7. Network Software

Programs used to manage networks.

Types

Network Operating System (NOS)

Examples:

- Windows Server
- Linux Server

Protocol Software

Examples:

- TCP/IP
- HTTP
- FTP

Security Software

Examples:

- Firewall
 - Antivirus
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8. Design Issues of Layers

Important Design Issues

Addressing

Identification of devices.

Error Control

Detecting and correcting errors.

Flow Control

Controls speed of communication.

Routing

Selects best path.

Fragmentation

Breaks large data into packets.

Connection Control

Reliable communication handling.

Multiplexing

Sharing communication channels.

Security

Protection from unauthorized access.

Synchronization

Maintains timing coordination.

9. OSI Reference Model

OSI = Open Systems Interconnection

OSI Layers

Layer	Function
Application	User services
Presentation	Encryption, compression
Session	Session management
Transport	Reliable delivery
Network	Routing
Data Link	Framing and error handling
Physical	Signal transmission

Characteristics

- 7-layer model
 - Standardized architecture
 - Easy troubleshooting
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10. TCP/IP Model

TCP/IP = Transmission Control Protocol / Internet Protocol

Layers

Layer	Function
Application	User services
Transport	Reliable communication
Internet	Routing and IP addressing
Network Access	Physical transmission

Characteristics

- Practical model
 - Used on Internet
 - Flexible architecture
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11. OSI vs TCP/IP

OSI	TCP/IP
7 layers	4 layers
Theoretical	Practical
Developed by ISO	Developed by DoD
Strict layering	Flexible layering

12. Physical Layer

Lowest layer of OSI model.

Functions

- Data transmission
- Signal conversion
- Bit synchronization
- Media handling

13. Network Devices

Hub

Connects devices and broadcasts data.

Repeater

Boosts weak signals.

Modem

Provides Internet connectivity.

Cables

- Twisted Pair
- Coaxial
- Optical Fiber

14. Network Topology

Topology = Arrangement of devices in network.

Types

Bus Topology

Single cable connection.

Star Topology

All devices connected to central hub.

Ring Topology

Circular arrangement.

Mesh Topology

Every device connected to others.

Tree Topology

Hierarchical arrangement.

Comparison

Topology	Advantage	Disadvantage
Bus	Cheap	Cable failure affects network
Star	Reliable	Hub failure affects network
Ring	Organized	One failure affects network
Mesh	Highly reliable	Expensive
Tree	Scalable	Complex

15. Transmission Media

Transmission media = Path for data transfer.

Guided Media (Wired)

Twisted Pair Cable

Cheap and commonly used.

Coaxial Cable

Better speed and less noise.

Optical Fiber

Very high speed and secure.

Unguided Media (Wireless)

Radio Waves

Used in Wi-Fi and radio.

Microwaves

Used in mobile communication.

Infrared

Used in TV remotes.

Satellite Communication

Long-distance communication.

16. Signal Transmission

Analog Signal

Continuous signal.

Examples:

- Voice
- Radio

Digital Signal

Binary signal (0 and 1).

Examples:

- Computer data

Analog vs Digital

Analog	Digital
Continuous	Discrete
More noise	Less noise
Less reliable	More reliable

17. Network Performance

Factors

Bandwidth

Maximum data carrying capacity.

Throughput

Actual successful data transfer.

Latency

Time delay in transmission.

Jitter

Variation in delay.

Reliability

Stability of network.

Formula

Important Formulas

1. Transmission Time

Transmission Time = Data Size / Bandwidth

2. Propagation Delay

Propagation Delay = Distance / Propagation Speed

3. Throughput

Throughput = Total Data Transferred / Time Taken

4. Bandwidth-Delay Product

Bandwidth × Delay

Represents amount of data present in the network at a time.

5. Total Delay

Total Delay = Transmission Delay + Propagation Delay + Processing Delay + Queuing Delay

18. Transmission Impairments

Problems affecting signal quality.

Types

Attenuation

Signal becomes weak.

Distortion

Signal shape changes.

Noise

Unwanted signals disturb communication.

SNR Formula

Important Formulas

1. Signal-to-Noise Ratio (SNR)

SNR = Signal Power / Noise Power

2. SNR in Decibel dB

$SNR_{dB} = 10 \log_{10} \text{SignalPower} / \text{NoisePower}$

3. Attenuation

$$\text{Attenuation } dB = 10 \log_{10} P_2/P_1$$

Where:

- P1 = Input Power
 - P2 = Output Power
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19. Public Switched Telephone Network (PSTN)

Traditional worldwide telephone network.

Components

- Telephone set
- Local loop
- Switching office
- Trunk lines

Features

- Uses circuit switching
 - Reliable voice communication
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20. Switching Techniques

(a) Circuit Switching

Dedicated communication path.

Example:

- Telephone network

(b) Message Switching

Complete message forwarded.

(c) Packet Switching

Message divided into packets.

Example:

- Internet

Comparison

Technique	Feature	Example
Circuit Switching	Dedicated path	Telephone
Message Switching	Store and forward	Telegraph
Packet Switching	Packet-based transfer	Internet

Important One-Line Definitions

Computer Network

A collection of interconnected computers that share resources and information.

LAN

A network covering a small geographical area.

MAN

A network covering a city or metropolitan area.

WAN

A network covering large geographical areas.

Internet

A worldwide network connecting millions of computers.

Network Topology

Arrangement of devices in a network.

Transmission Media

Path through which data travels.

Switching Techniques

Methods used to transfer data from source to destination.

Quick Memory Tips

- LAN → Small Area
- MAN → City Area
- WAN → Worldwide
- Hub broadcasts everywhere
- Switch sends to specific device
- Router connects networks
- OSI = 7 layers
- TCP/IP = 4 layers
- Optical Fiber = Fastest medium
- Packet Switching = Internet